sars — Extracted Transmission Models

Overview

A total of 157 transmission models were extracted from 80 articles that were flagged for containing extractable transmission models. The predominant model architecture was Compartmental (46.5%), with Airborne or close contact (48.4%) representing the most commonly modelled transmission route.

Summary

Metric	Value
Models extracted	157
Articles considered	80
Deterministic models	56 (35.7%)
Stochastic models	87 (55.4%)
Models with available code	10 (6.4%)

Figure 1: Model Overview

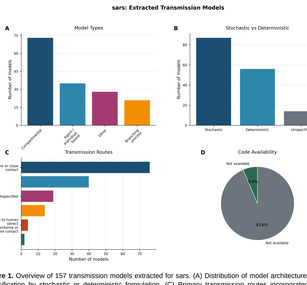

Figure 1. Overview of 157 transmission models extracted for sars. (A) Distribution of model architectures. (B) Classification by stochastic or deterministic formulation. (C) Primary transmission routes incorporated. (D) Proportion of models with publicly available source code.

Model Formulation

Deterministic formulations were employed in 35.7% of models, while stochastic approaches accounted for 55.4%. Empirical data were incorporated in 8.3% of the extracted models, with 46.5% classified as purely theoretical.

Figure 2: Interventions and Assumptions

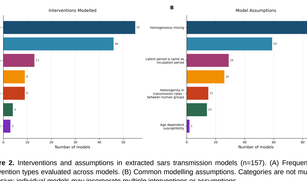

Figure 2. Interventions and assumptions in extracted sars transmission models (n=157). (A) Frequency of intervention types evaluated across models. (B) Common modelling assumptions. Categories are not mutually exclusive; individual models may incorporate multiple interventions or assumptions.

Interventions and Assumptions

The most frequently modelled interventions were Quarantine (35.0%), Behaviour changes (29.3%), Hospitals (8.3%). Common modelling assumptions included Homogeneous mixing (58.0%), Latent period is same as incubation period (18.5%), Heterogeneity in transmission rates - over time (16.6%).

Figure 3: Programming Languages

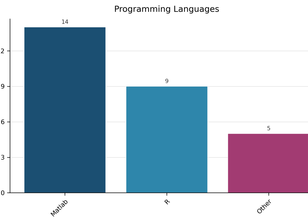

Figure 3. Programming languages used for model implementation among 28 models with specified language.

Figure 4: Model Type by Formulation

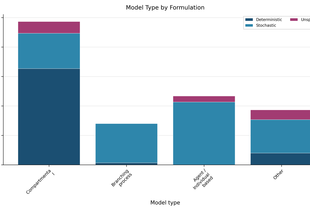

Figure 4. Distribution of deterministic and stochastic formulations across model types (n=157).

Reproducibility

Source code was publicly available for 6.4% of the extracted models.

Tables

Table 1: Model Types

Model Type	Count	Proportion
Compartmental	73	46.5%
Agent / Individual based	35	22.3%
Other	28	17.8%
Branching process	21	13.4%

Distribution of model architectures across 157 extracted transmission models.

Table 2: Model Formulation

Formulation	Count	Proportion
Stochastic	87	55.4%
Deterministic	56	35.7%
Unspecified	14	8.9%

Classification of extracted models by stochastic versus deterministic formulation (n=157).

Table 3: Transmission Routes

Transmission Route	Count	Proportion
Airborne or close contact	76	48.4%
Human to human (direct contact)	40	25.5%
Unspecified	19	12.1%
Airborne or close contact;Human to human (direct contact)	14	8.9%
Human to human (direct contact);Airborne or close contact	4	2.5%
Airborne or close contact;Unspecified	2	1.3%
Human to human (direct non-sexual contact);Airborne or close contact	1	0.6%
Human to human (direct non-sexual contact)	1	0.6%

Primary transmission routes incorporated in extracted models (n=157).

Table 4: Interventions Modelled

Intervention Type	Count	Proportion
Quarantine	55	35.0%
Behaviour changes	46	29.3%
Hospitals	13	8.3%
Contact tracing	9	5.7%
Vaccination	9	5.7%
Other	4	2.5%
Treatment	3	1.9%

Types of interventions evaluated in extracted models. Categories are not mutually exclusive (n=157).

Table 5: Model Assumptions

Assumption	Count	Proportion
Homogeneous mixing	91	58.0%
Other	59	37.6%
Latent period is same as incubation period	29	18.5%
Heterogeneity in transmission rates - over time	26	16.6%
Heterogeneity in transmission rates - between human groups	15	9.6%
Heterogeneity in transmission rates - between groups	14	8.9%
Age dependent susceptibility	2	1.3%

Common modelling assumptions. Categories are not mutually exclusive (n=157).

Table 6: Code Availability

Code Available	Count	Proportion
Yes	10	6.4%
No	146	93.0%

Availability of publicly accessible source code for extracted models (n=157).

Table 7: Programming Languages

Programming Language	Count	Proportion
Unspecified	129	82.2%
Matlab	14	8.9%
R	9	5.7%
Other	5	3.2%

Programming languages used for model implementation (n=157).

Table 8: Empirical Data Usage

Empirical Data Used	Count	Proportion
Yes	13	8.3%
No/Unspecified	144	91.7%

Model Inventory (Sample)

Table 9 presents a sample of 10 extracted model records. The complete dataset of 157 extracted transmission models for sars is publicly available.

Table 9: Sample of Extracted Models

Article ID	Model Type	Compartmental Structure	Formulation	Transmission Route	Spatial Sc
PMID_12766207	Compartmental	SEIR	Deterministic	Airborne or close contact	Unspecifie
PMID_12766207	Branching process	Not compartmental	Stochastic	Airborne or close contact	Unspecifie
PMID_15353409	Agent / Individual based	Not compartmental	Stochastic	Airborne or close contact	Unspecifie
PMID_15353409	Other	Not compartmental	Deterministic	Airborne or close contact	Unspecifie
PMID_15498594	Agent / Individual based	Not compartmental	Unspecified	Airborne or close contact	Unspecifie
PMID_15498594	Agent / Individual based	Not compartmental	Unspecified	Airborne or close contact	Unspecifie
PMID_15498594	Agent / Individual based	Not compartmental	Unspecified	Airborne or close contact	Unspecifie
PMID_15306395	Compartmental	SIR	Deterministic	Human to human (direct contact)	Unspecifie
PMID_15306395	Compartmental	Other compartmental	Stochastic	Human to human (direct contact)	True
PMID_15306395	Branching process	Not compartmental	Stochastic	Human to human (direct contact)	Unspecifie

Sample of 10 records from 157 extracted sars transmission models.

Data Availability

The complete dataset of extracted transmission models for sars (n=157) has been made publicly available to support future modelling efforts and outbreak preparedness.